Alexander Chin

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Education

University of California San Diego **Sep 2025 – Jun 2028**
B.S. in Mathematics-Computer Science, Minor in Cognitive Science *GPA: 3.7/4.0*

Technical Skills

Languages: Python, TypeScript/JavaScript, Java, C++, Go, SQL

Frameworks: React, FastAPI, REST, WebSockets, Playwright, PyTorch, pandas

Data & Infra: PostgreSQL, MongoDB, AWS (Lightsail, S3), Docker, Linux, cron, Git/GitHub

Coursework: Data Structures & Algorithms, Systems Programming, Graph Theory, Machine Learning, Convex Optimization

Experience

IBM **Lowell, MA**
Incoming Software Developer Co-op *Spring 2027*

UCSD Visual Learning Lab **La Jolla, CA**
Undergraduate Research Assistant (PI: Prof. Bria Long) *Mar 2026 – Present*

- Fixed race conditions in the lab's gaze-annotation tool by centralizing claim and lock files and replacing PID-based temp file names with UUIDs, letting annotators on separate machines work concurrently.
- Extended the lab's open-source embedding package (vllpy), used across 4 lab projects, with 12 commits adding ranked RDM analysis, a BabyView-trained DINOv3 model, and a transformers compatibility fix.
- Streamlined gaze coding for 120+ participants and 7+ GB of video by routing iCatcher+ output to the lab server with per-subject validation checks, and wrote setup and troubleshooting docs.

Association for Computing Machinery (ACM) at UCSD **La Jolla, CA**
Projects Director; previously AI Projects Mentor *Mar 2026 – Present*

- Led a 6-person team building a multiplayer drawing game scored by computer vision, assigning ML, backend, and frontend work, reviewing pull requests, and integrating 4 PyTorch CNNs into 1 inference pipeline.
- Designed the game-state schema and 3-tier ranked scoring, and replaced softmax confidence with class-centroid cosine similarity.
- Run the project cycle for 8 SWE/AI/Design teams via standups; organized workshops and showcases drawing 250+ students.

Essenergy Inc. **Milpitas, CA**
Software Engineer, Contract *Jun 2026 – Sep 2026*

- Proposed and shipped end-to-end order-processing automation in Python/Playwright covering authentication, labels, tracking, invoicing, and inventory for 30+ weekly orders across 3 vendor portals.
- Made daily runs fully unattended by deploying the Dockerized workflow on AWS Lightsail with cron, screenshot-and-log email alerts on failure, and a CLI resume flag to recover interrupted runs.
- Cut daily remittance reconciliation from ∼30 min to under 1 min with a pipeline that auto-matches invoice numbers across 1,800+ rows from 2 vendors and flags invalid or suspicious entries.

IDX Exchange **Remote**
Data Analyst Intern *Jun 2026 – Sep 2026*

- Built a reproducible Python pipeline for 400,000+ CRMLS real-estate records automating ingestion, monthly consolidation, validation, feature engineering, and IQR outlier removal, dropping ∼11% unusable records.
- Identified Sacramento as a high-growth market via analysis-ready feature tables and 2 Tableau dashboards covering closing activity, geographic trends, and agent and office performance.

Projects

TSS++ | *React, TypeScript, FastAPI, MongoDB* *Open source, 200+ users*

- Launched an open-source course planner as an alternative to UCSD's new student system (100+ commits).
- Built prerequisite discovery across 6,000+ courses by reverse-engineering the undocumented TSS OData API from network traffic and modeling prerequisites as an AST with BFS/DFS traversal, depth limits, and cycle detection.

SafeStreets | *Claude, multi-agent* *UC Berkeley AI Hackathon 2026*

- Won Best Use of Claude among 200+ projects and 1,000+ hackers (team of 4) with a multi-agent system that turns street imagery and public records into ready-to-send city council letters.
- Co-built a two-stage vision pipeline: Claude assesses 1 satellite and 4 Street View images blind to community reports, then a separate pass corroborates each hazard against crash, 311, and news records.

SwapU | *FastAPI, Supabase, WebSockets, Fetch.ai*

- Built the FastAPI backend (REST, WebSockets, Supabase PostgreSQL, S3, auth) for a student item-swap marketplace.
- Integrated BrowserUse agents that retrieve real-time market prices and confidence scores so Fetch.ai agents can negotiate multi-item trades, and validated the negotiation flow against 128 mock listings.